

Relationship Types & Relationship Sets

Q. Explain the concepts of Relationship Types and Relationship Sets. Discuss the Degree of Relationship and Mapping Cardinalities (1:1, 1:N, M:N) with examples.

[DEF] Definition to Remember

A Relationship Type defines a meaningful association between two or more entity types, represented by a Diamond in ER diagrams. A Relationship Set is the actual collection of specific relationship instances existing in the database at any given moment.

1. Relationship Type and Relationship Set

* Relationship Type: Association between entity types. E.g., WORKS_FOR links EMPLOYEE and DEPARTMENT.

2. Degree of Relationship

The Degree = number of entity types participating in the relationship.

Degree	Name	Example
1	Unary	Employee `Manages` another Employee
2	Binary	Student `Enrolls_In` Course (most comm
3	Ternary	Doctor, Patient, Drug connected by `Pr

3. Mapping Cardinalities

Mapping Cardinality defines the maximum number of relationship instances an entity can participate in.

1. One-to-One (1:1)

2. One-to-Many (1:N)

3. Many-to-One (N:1)

4. Many-to-Many (M:N)

1:1	->	A	<R>	B
1:N	->	A	<R>	B
M:N	->	A	<R>	B

4. Participation Constraints

Type	Description	ER Symbol
Total Participation	Every entity in the set MUST participa	Double Line ()
Partial Participation	Entity may or may not participate	Single Line ()

* Total Example: Every EMPLOYEE must belong to a DEPARTMENT.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Relationship Type = association between entity types (Diamond); Relationship Set = collection of all instances.

2. Degree: Unary (1), Binary (2 -- most common), Ternary (3).
3. Mapping Cardinalities: 1:1, 1:N, N:1, M:N -- define how many entities can participate.
4. 1:1: one entity maps to exactly one (e.g., Manager -> Department).
5. 1:N: one on one side, many on the other (e.g., Department -> Employees).
6. M:N: many on both sides -- requires a Junction Table in the relational model.
7. Total Participation (double line): every entity MUST participate; Partial (single line): optional.

[RECALL] Quick Recall

Relationship Type (Diamond) -> Degree (Unary/Binary/Ternary) -> Cardinality (1:1, 1:N, M:N) -> Participation (Total=double line, Partial=single line)